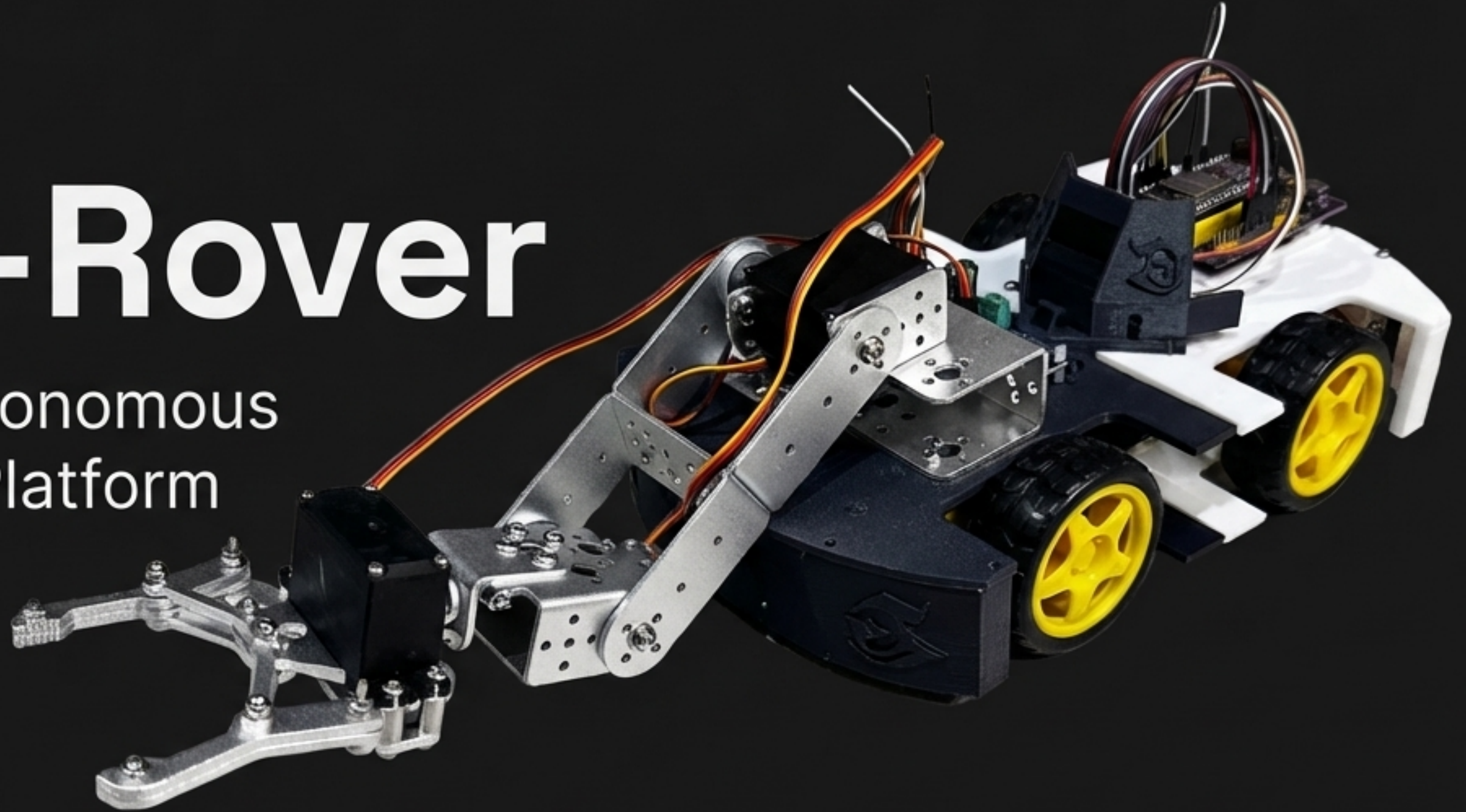


```
motors.begin();  
display.begin();  
xbox.begin();  
track.begin();  
contry.begin();  
smart.begin();
```

The BT-Rover

ESP32-Powered Autonomous
& Manual Robotics Platform

```
xbox.begin();  
motors.begin();  
motors.begin();  
motors.begin();  
bulloon.begin();  
track.missplar.begin();  
track.ny.begin();  
vm.setup();
```



Platform Capabilities

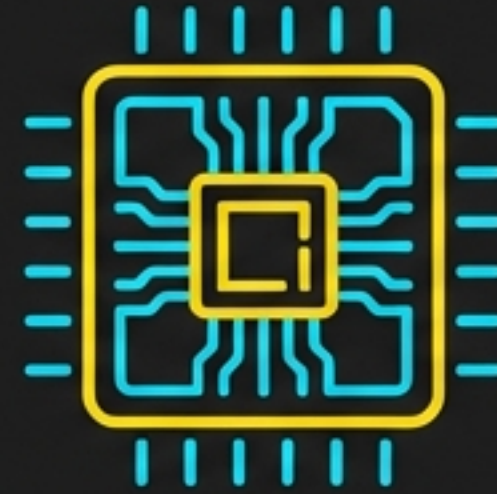


Teleoperation

Xbox Control via Bluepad32

Seamless Bluetooth pairing and arcade-style dual-stick driving.

```
BLE::pair("Xbox Controller");  
drive(STICK_L, STICK_R);
```

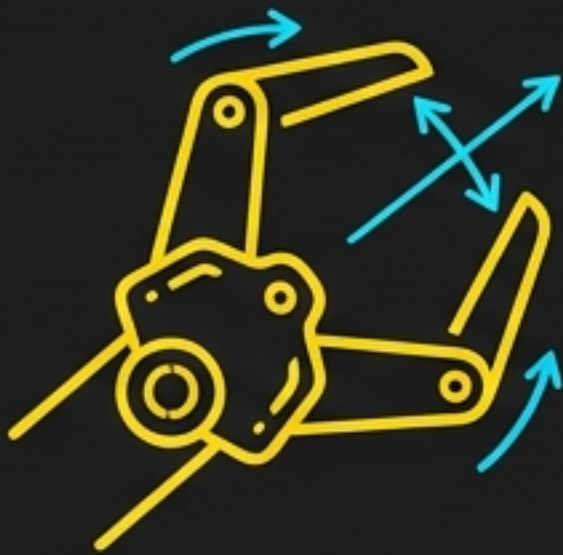


Central Compute

ESP32 Core

The WROOM-32 module acts as the brain, running a heavily modular object-oriented C++ stack.

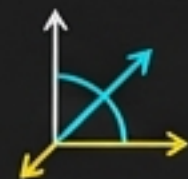
```
NCU: ESP32-WROOM-32; CORE: Dual  
240MHz; ARCH: OOCPP/FreeRTOS
```



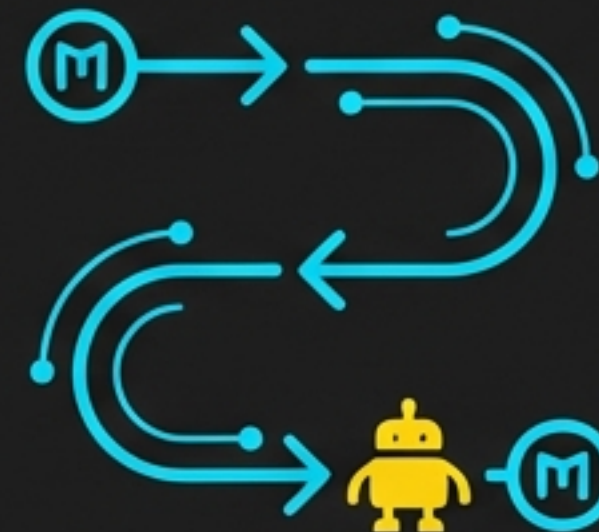
Manipulation

3-DOF Robotic Arm

A modular servo-driven metallic arm with custom kinematics for parallel gripping.



```
JOINT_ANGLES: {θ1: 30°,  
θ2: -45°, θ3: 10°};  
GRIPPER: PARALLEL
```



Autonomy

Flash-Memory Trajectory

Records physical micro-movements into Non-Volatile Storage (NVS) for headless playback.

```
NVS::record("traj_001.dat");  
NVS::playback("traj_001.dat", LOOP);
```

Hardware Architecture: The Anatomy

Node A: The Brain

ESP32 Dev Board

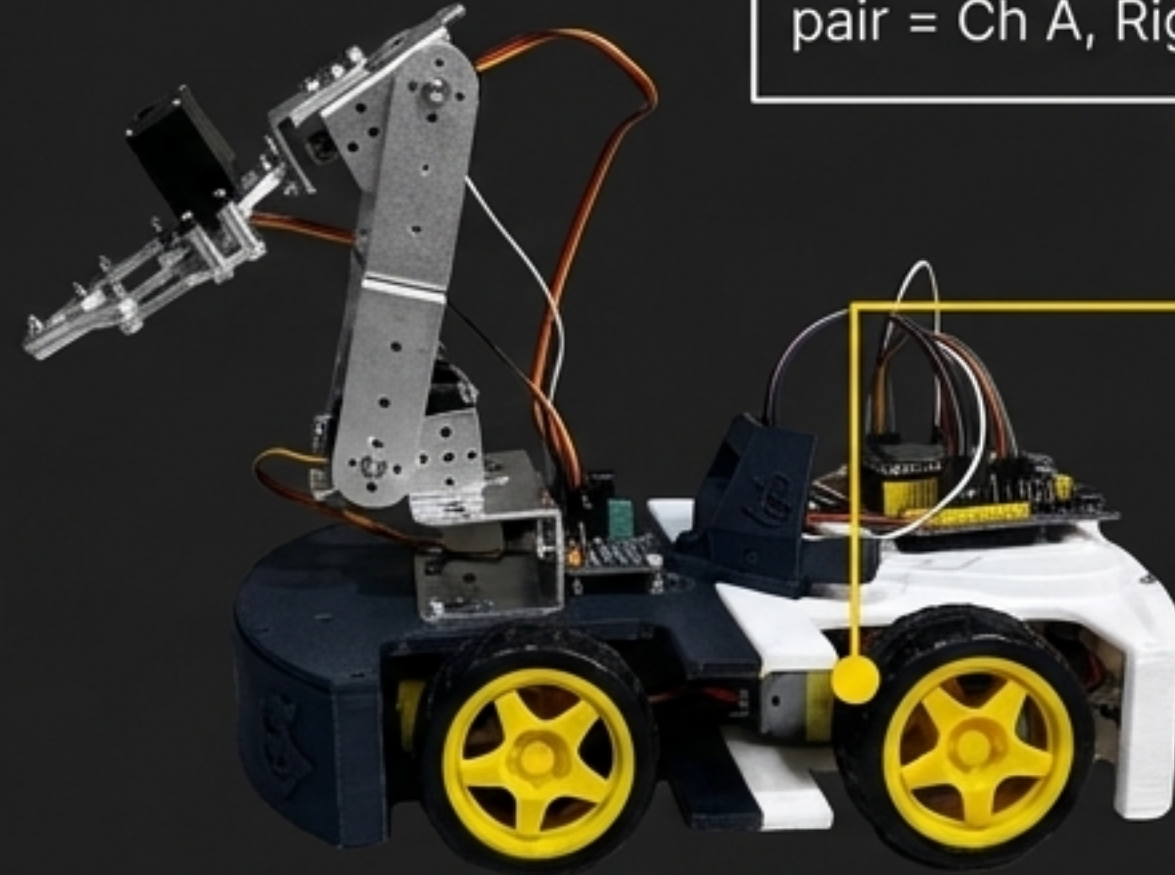


Node C: The Power

LM2596 Buck Converters.

Node B: The Brawn

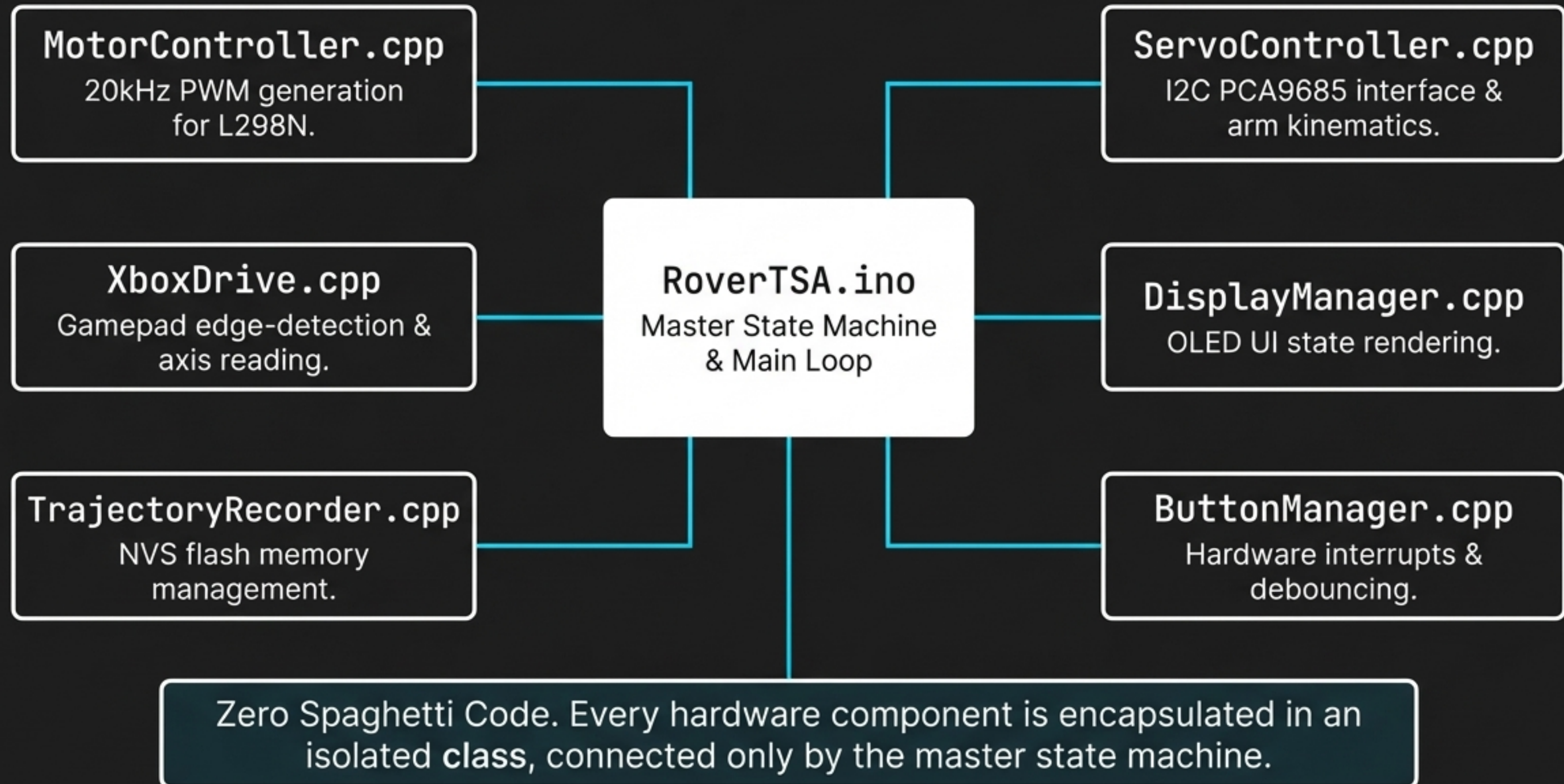
L298N Motor Driver. Wired in parallel for 4 DC motors (Left pair = Ch A, Right pair = Ch B).



Node D: The Precision

PCA9685 I2C Servo Driver.

Software Modularity: The Nervous System

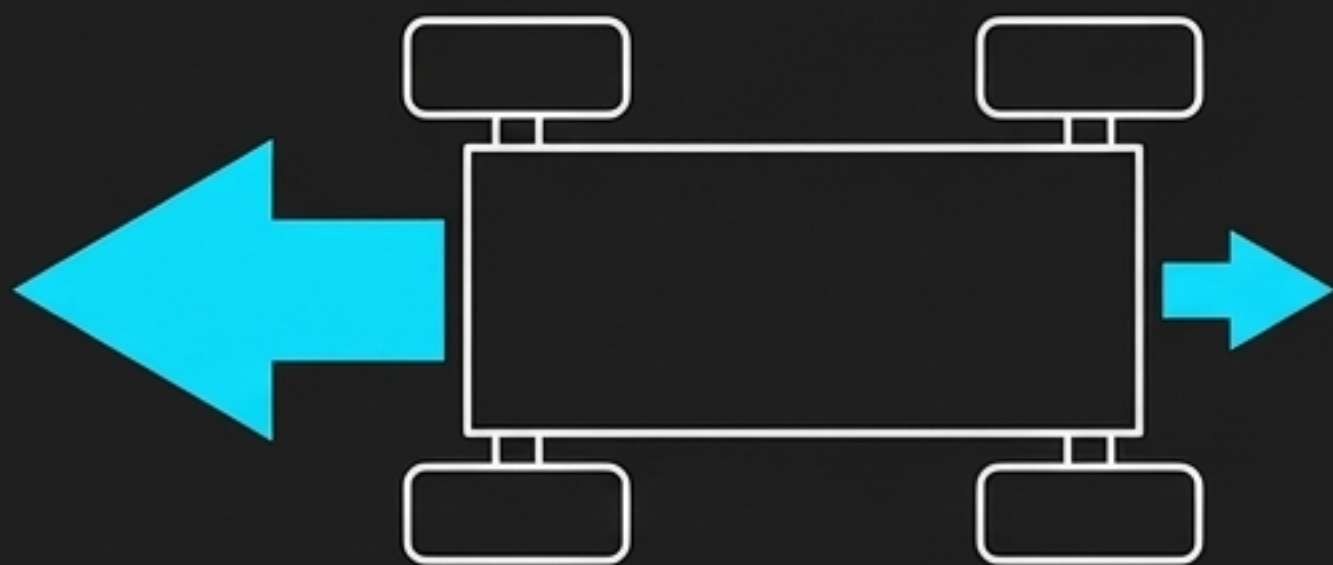


Locomotion & Drivetrain

Arcade Mixing Diagram



left = throttle + steer
right = throttle - steer

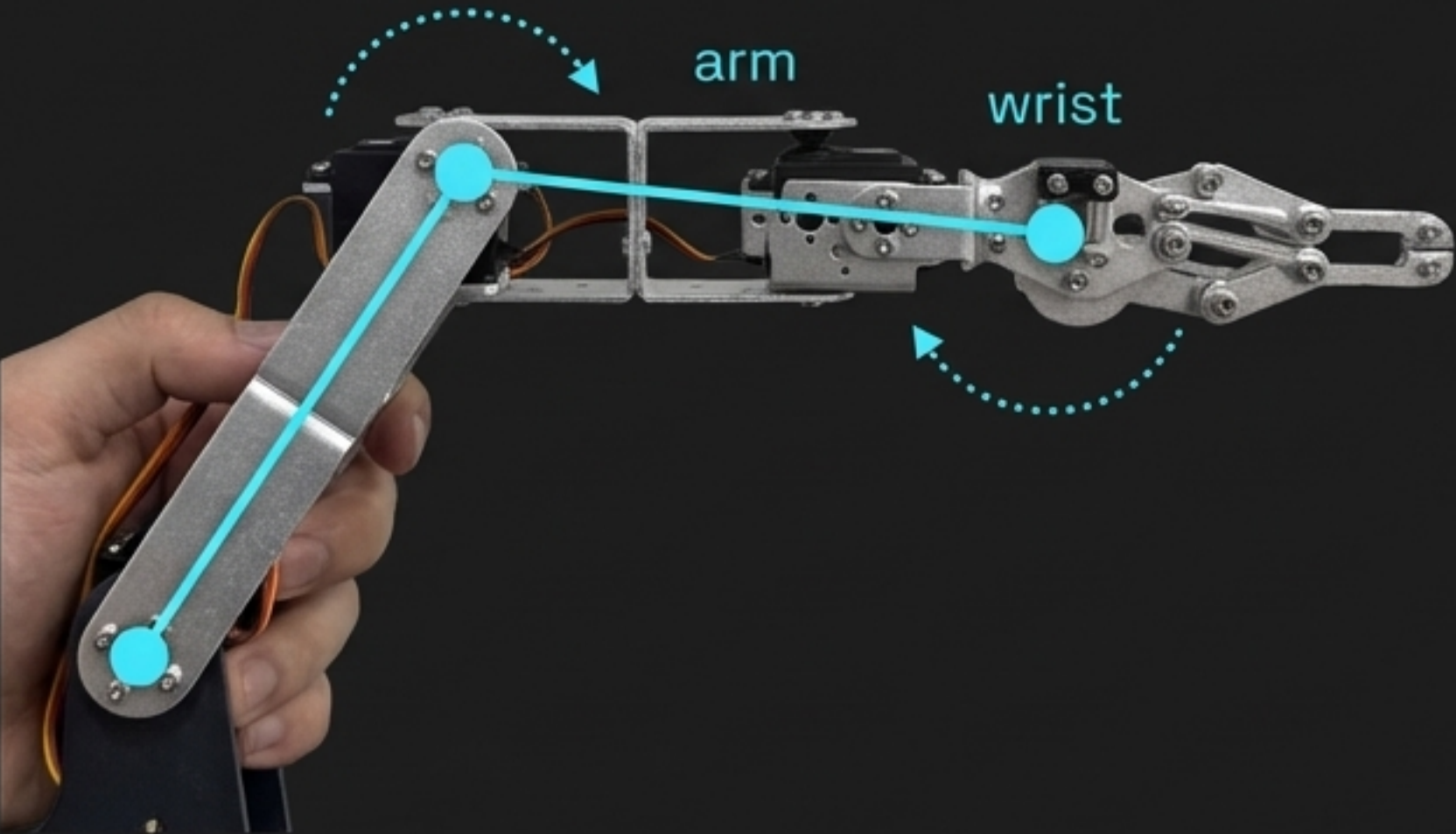


Speed Mode Comparison Matrix

Speed Mode	Speed Scale Float	Min PWM Clamp	Ideal Use Case
Fast	1.0f	70	Open Navigation
Slow	0.75f	90	Precision Arm Manipulation

Manipulation: Kinematic Logic

The Sync Mode Kinematic Model



`wrist = wristDefault - (arm - armDefault) + trim`

Arm Mode Diagnostic Table

Dimension	Free Mode	Sync Mode
User Input	Direct joystick axis mapping	Wrist trim mapping
Kinematic Logic	Independent joint angles	Computed wrist angle based on shoulder delta
Ideal Use Case	Custom arbitrary positioning	Level-carrying tasks (e.g. holding liquid)

Interface & Telemetry

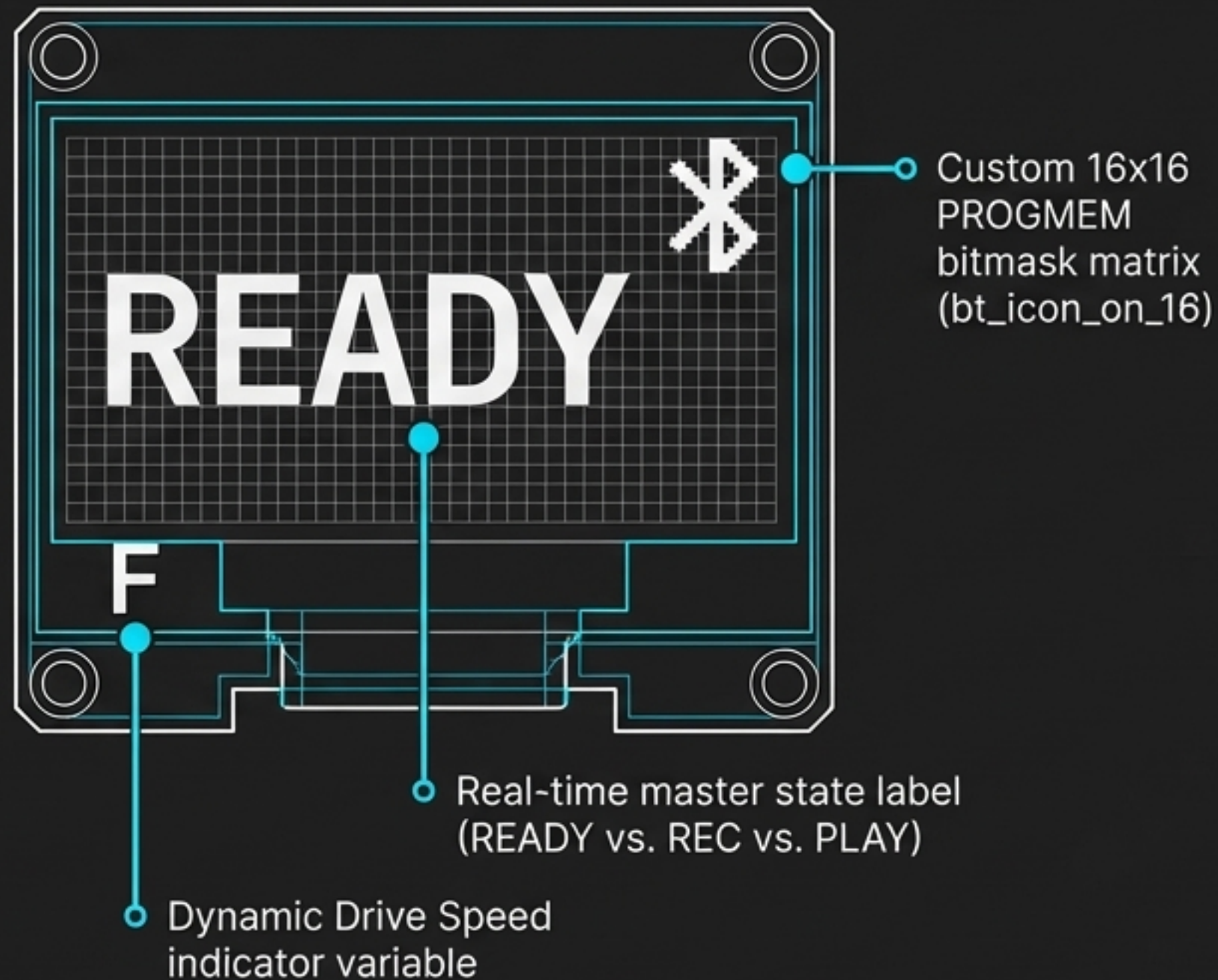
Input Processing (XboxDrive.cpp)



```
uint32_t misc = gp->miscButtons();  
bool nowDown = (misc & 0x00000004) != 0;
```

Bitwise masking guarantees microsecond-accurate hardware interrupts without blocking the main drive loop.

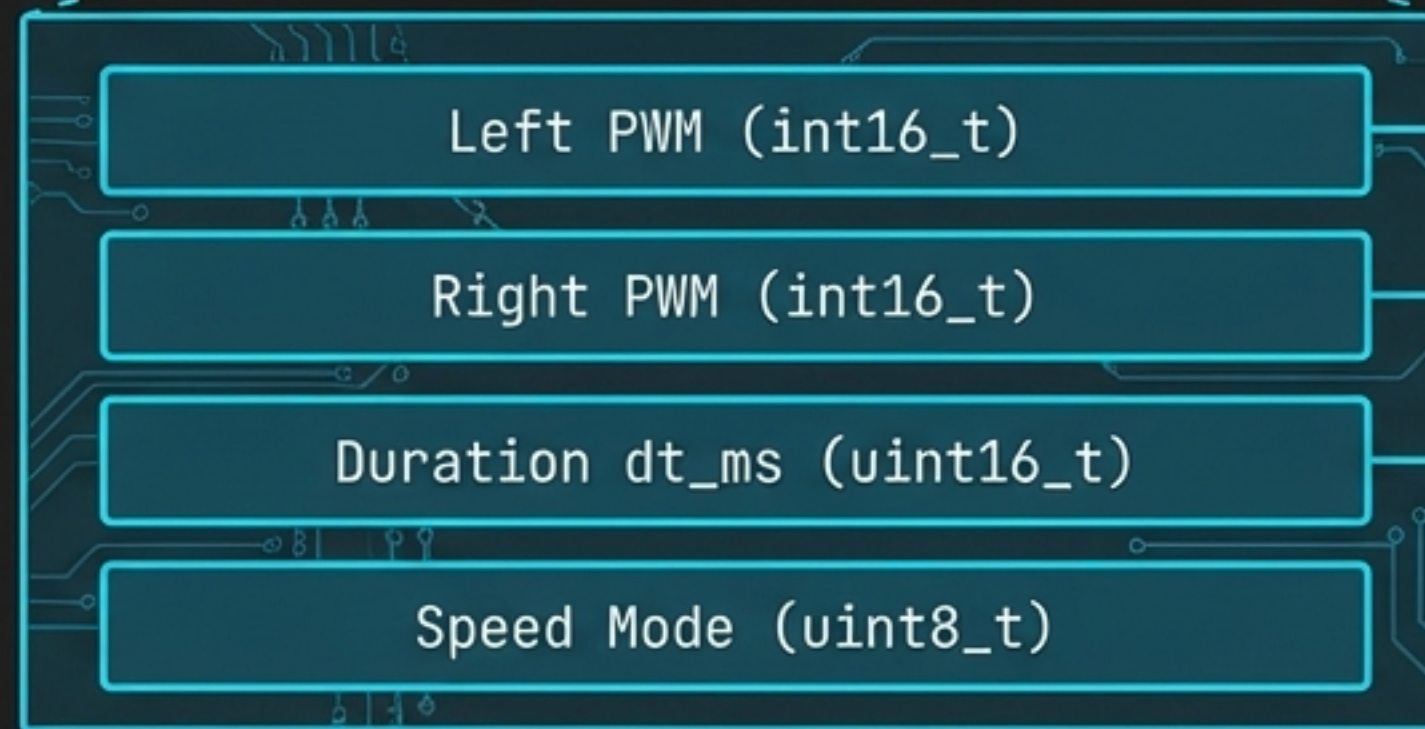
Telemetry Rendering (DisplayManager.cpp)



Killer Feature: Trajectory Memory

Flash Memory Buffer Array

Segment `_buf[MAX_SEGMENTS = 600]`



Anatomy of a Segment

By storing elapsed time (`dt_ms`) instead of discrete ticks, the buffer drastically reduces its memory footprint. This architecture enables minutes of complex driving to be compacted and saved directly to the ESP32's internal Non-Volatile Storage (NVS).

Failsafes & State Management

Warning

The busyBefore Block

Prevents the UI from accidentally opening the menu during the exact loop iteration when a user ends a recording. Strict state isolation.

Status Screen
(Drive/Record/Play)

Menu Screen
(Settings)

Warning

Hardware Watchdogs

If the Bluetooth connection drops mid-drive, the state is instantly forced back to Idle.

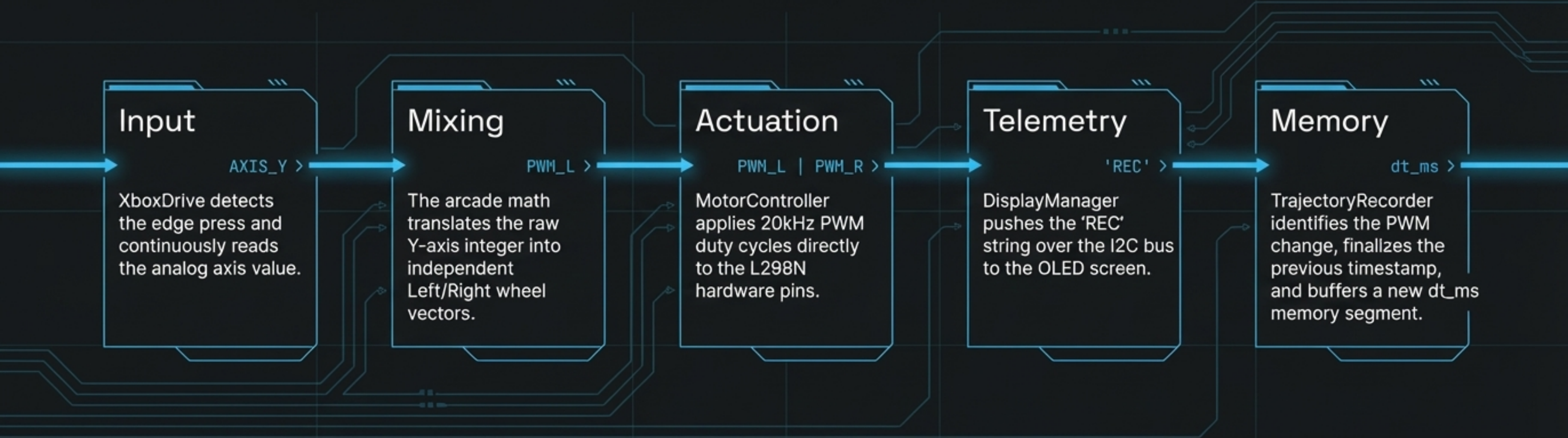
Warning

Global Stops

Physical long-presses or Xbox menu buttons instantly trigger `motors.stop()` overriding any active trajectory playback.

System Synthesis: The Data Pipeline

The Event: User pushes the left thumbstick forward while pressing 'Record'.



A Perfectly Integrated Orchestra. Modules operate independently but execute a unified, microsecond-perfect physical action.

Future Roadmap: Built to Scale

Spatial Awareness



Obstacle Avoidance

Integrating ultrasonic or LiDAR sensors into the main loop, creating a new decoupled `SensorController.cpp` module to override manual locomotion when proximity limits are breached.



Integrating ultrasonic or LiDAR sensors into the main loop, creating a new decoupled `SensorController.cpp` module to override manual locomotion when proximity limits are breached.

Infinite Memory



SD Card Storage

Bypassing the 600-segment `ESP32` internal `NVS` limit by utilizing the high-speed `SPI` bus to read and write multiple complex, multi-minute trajectory files to removable media.



Bypassing the 600-segment `ESP32` internal `NVS` limit by utilizing the high-speed `SPI` bus to read and write multiple complex, multi-minute trajectory files to removable media.

Advanced Manipulation



Inverse Kinematics

Upgrading the current 2D `computeSyncWristDeg` logic to a full 3D Cartesian `XYZ` spatial solver, enabling precise coordinate-based gripping and waypoint pathing for the arm.



Upgrading the current 2D `computeSyncWristDeg` logic to a full 3D Cartesian `XYZ` spatial solver, enabling coordinate-based gripping and waypoint pathing for the arm.